

DAILY CLASS NOTES

BPSC GENERAL SCIENCE

Lecture – 03 CELL PART 2



CELL PART 2

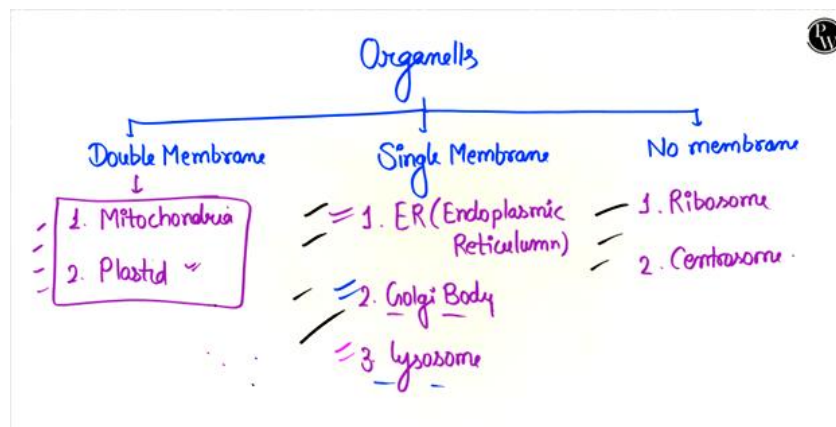
Cell Organelles

- ❖ The cellular components are called cell organelles. These cell organelles include both membrane and non-membrane bound organelles, present within the cells and are distinct in their structures and functions.
- ❖ They coordinate and function efficiently for the normal functioning of the cell. A few of them function by providing shape and support, whereas some are involved in the locomotion and reproduction of a cell.
- ❖ There are various organelles present within the cell and are classified into three categories based on the presence or absence of membrane.

Organelles without membrane: The Cell wall, Ribosomes, and Cytoskeleton are non-membrane-bound cell organelles. They are present both in the prokaryotic cell and the eukaryotic cell.

Single membrane-bound organelles: Vacuole, Lysosome, Golgi Apparatus, Endoplasmic Reticulum are single membrane-bound organelles present only in a eukaryotic cell.

Double membrane-bound organelles: Nucleus, mitochondria and chloroplast are double membrane-bound organelles present only in a eukaryotic cell.



- ❖ **Endoplasmic reticulum :-**
 - It is a single membrane structure.
 - It is present in both plant and animal cells.
 - It is of two types.



❖ **Rough endoplasmic reticulum :-**

- Ribosomes are found in it.
- It synthesizes proteins.

❖ **Soft/smooth endoplasmic reticulum :-**

- Ribosomes are not found in it.
- It synthesizes fats.

❖ **Golgi Body :-**

- This is also a single membrane structure.
- It is a group of flat and folded sacs of different sizes which are spread between the endoplasmic reticulum and the cytoplasmic layer.
- It works to transport, analyze, pack, etc. the proteins made by the endoplasmic reticulum.
- It is also known as the traffic police of cell.
- Apart from blue green algae, they are found in the cells of all other living organisms. In plants they are also called dictyosomes.
- **Functions :-**
 - ✓ Various enzymes are secreted and stored by them.
 - ✓ Primary lysosomes are produced by
 - ✓ Cell walls are formed in plant cells by these.
 - ✓ Golgi bodies are helpful in the secretion of hormones in animal cells.

❖ **Lysosome :-**

- It is found only in animals.
- Generally the shape of lysosome is circular or irregular and its structure is about 0.2 μ m to 0.8 μ m in diameter.
- These are sacs surrounded by layered membranes.
- They are full of digestive enzymes.
- It is also known as cell suicide bag, digestive bag and atom bomb.
- Hydrolytic enzymes are found in it.

- Function:- It destroys infected cells and dead cells.
- It is found most in T-lymphocytes of WBC.

❖ **Ribosome :-**

- It is a cell without membrane.
- It is present in both plant and animal cells.
- This is the smallest organelle.
- It is made up of RNA and protein (Ribonucleoprotein).
- It is 80s and 70s type.
- S stands for Sedberg unit.
- 70s are found in prokaryotic and 80s eukaryotic cells.
- George Pallad (1953) was the first to observe the dense granular structure ribosome through electron microscope. These are made of ribonucleic acid and protein and are not surrounded by any membrane.
- Eukaryotic ribosomes are of 80s type and prokaryotic ribosomes are of 70s type. Each ribosome has a large subunit and a small subunit.
- This indirectly expresses the size and density. Both 70s and 80s ribosomes are made up of two subunits.

❖ **Centrosome :-**

- Its structure is spherical which is located almost in the middle on the outer surface of the nucleus.
- It is found only in animals.
- It is made of tubulin protein.
- Function :-
 - ✓ It helps in the formation of spindle fibers during cell division.
 - ✓ Due to this, at the time of cell division, they are classified into two groups and two opposite poles are formed.

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